

Marathon Runner Portal

US01: User Registration

- **As a new user, I want to** register with email, password, and personal details **so that** I can access the portal after verification.

Business Rules:

1. Email must be unique (no duplicate accounts).
2. Password must be at least 12 characters with 1 special character.
3. Users cannot log in until email is verified. Allowed domains (gmail and yahoo).
4. Users must be with age between 18 – 40.
5. Show “Account Created at: [timestamp]” in User Profile.
6. Provide status for user account (Pending / Verified)
7. If runner, then 500 marathon points will be credited in account.

US02: Login & Role-based Dashboard

- **As a user, I want to** log in and see a dashboard based on my role (Admin/Runner).

Business Rules:

1. After **5 failed attempts**, account locks for **30 minutes**.
2. Admins see analytics; runners see upcoming races and training stats.
3. Session expires after **1 hour of inactivity**.

US03: Race Listing and Filtering

- As a runner, I want to see available races with filters (date, location, distance).

Business Rules:

1. Only show races where registration is open.
2. Hide races that occur in <48 hours (registration closed).
3. Runners cannot see races they’ve already registered for.

US04: Race Registration

- As a runner, I want to register for a race only if I meet training requirements.

Business Rules:

1. Must have ≥ 5 training logs in the last 30 days.
2. Cannot register for two races on the same day.
3. Registration must be ≥ 48 hours before race start time.
4. Show “Last Modified: [timestamp]” if admin changes race details.

US05: Training Log Creation

- As a runner, I want to log my training with distance, duration, and date.

Business Rules:

1. No future dates allowed.
2. Max 100 km per entry (prevent typos).
3. Duplicate logs (same date & distance) are rejected.

4. Logs older than 6 months cannot be edited.
5. Show “Logged 3 days ago” recency indicator.
6. “Edited” badge if **modified_at** differs from **created_at**.

US06: Training Progress Dashboard

- As a runner I want to view my training progress in a sortable, filterable table, So that I can analyze my performance with precision.

Business Rules:

1. Show average pace, distance trends, and comparisons.
2. Allow export to CSV/PDF for personal records.

US07: Forum Post Submission & Moderation

- As a runner, I want to post in the forum, but it must be approved.

Business Rules:

1. Posts with banned words are auto-flagged.
2. Admins must approve before visibility (max 24-hour delay).
3. Users with 3+ flagged posts are temporarily banned.

US08: User Profile and Past Races

- As a runner, I want to see my profile with past race history.

Business Rules:

1. Only show completed races (not upcoming).
2. Display personal best times per race distance.
3. Allow profile edits except email & username & id.

US09: Admin Race Management

- As an admin, I want to add/edit/delete races with validation.

Business Rules:

1. Cannot delete a race if registrations exist.
2. Race dates must be in the future.
3. Distance must be 5K, 10K, Half, or Full Marathon.
4. Provide “Deactivate” button instead of “Delete”.
5. Provide the status for races (Active / Inactive).

US10: Admin Analytics Dashboard

- As an admin, I want to see stats on user activity, race participation and posts shared by the runners for approval.
- Create a filtered table to view the approved and non-approved posts of runners.

Business Rules:

1. Data should be exportable as CSV/Excel.
2. User(admin) must be authenticated before exporting the file.

US11: User Authentication

1. Log all the failed login activities for security audits.
2. Provide error message along with required Http status code to front end if user data is not found in the database.

US12: Race CRUD with Validation

1. Race dates must be ≥ 7 days in the future.
2. Admins cannot modify races < 24 hours before start time.
3. Race names must be unique per year.

US13: Race Registration Validation

1. Reject if user has no training logs in last 30 days.
2. Prevent registration if race is at full capacity.
3. Charge 50 extra marathon points if registering within 7 days of race.

US14: Training Log Validation & Duplicate Check

1. Logs must have distance ≥ 1 km and ≤ 100 km.
2. Reject if pace is unrealistic (< 3 min/km or > 10 min/km).
3. Auto-flag logs that are 10x higher than user's average.

US15: Forum Moderation Workflow

1. Posts with links require manual approval.
2. Auto-delete posts older than 1 year.

US16: Email Service for Verification

1. Retry failed emails 3 times before marking as failed.

US17: Analytics Data Aggregation

1. Compute average training distance per user.
2. Compute Top 3 runners from the race.
3. Update the flag status of users who suddenly stop training (injury risk).
4. Compute the total registered runners from each race.

US18: Scheduled Job for Race Registration Closure

1. Close registration 48 hours before race start.
2. Auto-refund the marathon points if race is cancelled due to weather.

US19: Role-Based API Authorization

1. Block users(runners) from accessing `"/admin/*"` endpoints.
2. Log all admin actions for audit trails.

US20: Data Export for Admin

1. Generate CSV reports on Users, Races, Posts and Training logs.
2. Only admin can request exports.

Database Schema

1. Users (user_id, username, email, password, marathon_points, role, verified)
2. Races (race_id, name, date, distance, registration_open)
3. Registration (reg_id, user_id, race_id, reg_date)
4. TrainingLogs (log_id, user_id, distance, duration, date)
5. ForumPosts(post_id, user_id,